SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Preprocessing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:20**

Annexure A

CLASS TEST

Criteria	Understanding of Task ***(2)***	Procedure & Execution ***(2.5)***	Observation & Result ***(2)***	Viva Voce / Conceptual Response ***(2)***	Time Management & Presentation ***(1.5)***	***Total(10)***
Weightage	*20%*	25%	20%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

*I **J.Mohamed Aseem(192511258)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J.Mohamed Aseem
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS: **Total Marks: 10x2 =20**

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formals [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyse how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set x = {4,5,6,7,9}, y = {8,3,7,5,6}

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analysing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples and evaluate their impact on data analysis. [L3, L4, l5]

Answers

Q1.)

Data Mining: Data mining is the process of discovering useful patterns, relationships, and knowledge from large amounts of data.

Two basic data mining tasks:

1. Classification: Assigning data to predefined categories, such as classifying emails as spam or not spam.
2. Clustering: Grouping similar data items, such as grouping customers based on purchasing behaviour.

Steps involved in the KDD process:

1. Data Selection: Select relevant data from databases.
2. Data Preprocessing: Remove errors, handle missing values, and correct inconsistent data.
3. Data Transformation: Convert data into a suitable format for mining.
4. Data Mining: Apply algorithms to discover patterns.
5. Pattern Evaluation: Identify useful and meaningful patterns.
6. Knowledge Presentation: Present the discovered knowledge using reports, charts, or visualizations.

Role of data preprocessing: Data preprocessing improves data quality, reduces noise, handles missing values, and increases the reliability of data mining results. Without preprocessing, incorrect or incomplete data may produce misleading patterns.

Q2.)

Data cleaning is the process of identifying and correcting errors, missing values, inconsistencies, and noise in a dataset.

Two common data quality issues:

1. Missing values.
2. Noisy or incorrect data.

Methods to handle missing values:

- Deletion: Remove records with too many missing values. Example: remove a customer record when essential information is unavailable.
- Mean or median imputation: Replace missing numerical values with the mean or median. Example: replace a missing age with the median age of 25 years.
- Mode imputation: Replace a missing categorical value with the most frequent category. Example: replace a missing department with the most common department.

Methods to handle noisy data:

- Binning: Group sorted values into bins and smooth the values using bin means or medians.
- Regression: Use a fitted relationship to estimate appropriate values.
- Outlier treatment: Investigate extreme values and correct, cap, or remove them when justified.

Example: If the ages are 20, 21, 22, 200, the value 200 may be a data-entry error. It should be verified and corrected if possible.

Conclusion: Data cleaning improves accuracy, consistency, and reliability before analysis.

Q3.)

Data integration is the process of combining data from multiple sources into a single, consistent dataset.

Two challenges:

1. Data redundancy: The same information may appear in multiple sources.
2. Data inconsistency: Different sources may store the same information in different formats or with conflicting values.

Handling redundancy:

- Identify duplicate records using unique identifiers.
- Remove duplicate entries after verification.
- Use entity matching to identify records representing the same person or object.

Handling inconsistency:

- Standardize formats, such as converting dates to DD-MM-YYYY.
- Use consistent naming conventions, such as Male instead of M, male, and MALE.
- Resolve conflicting values using reliable source rules and validation.

Example: If two databases contain the same customer as C101 and C101, the records can be matched and duplicates removed. If one database stores Chennai as Chennai and another as Madras, a consistent naming convention should be applied.

Conclusion: Data integration creates a unified dataset by resolving duplication and inconsistencies, making analysis more reliable.

Q4.)

Covariance analysis is a statistical technique used to determine whether two numerical variables tend to change together.

- Positive covariance: Both variables tend to increase or decrease together.
- Negative covariance: One variable tends to increase when the other decreases.
- Zero covariance: There is no linear co-variation indicated by the covariance value.

Two covariance formulas:

1. Population covariance:

$$\operatorname{Cov}(X,Y)= \frac{\sum_{i=1}^{N}(X_i-\mu_X)(Y_i-\mu_Y)}{N}$$

2. Sample covariance:

$$s_{XY}= \frac{\sum_{i=1}^{n}(X_i-\bar X)(Y_i-\bar Y)}{n-1}$$

Here, μ_X,μ_Y are population means, while $\bar X,\bar Y$ are sample means.

Covariance helps identify relationships between variables and supports feature analysis in data mining.

Q5.)

Answer:

Covariance helps determine how the prices of two stocks change together.

Given:

- Stock A: 2, 3, 5, 4, 6

- Stock B: 5, 8, 10, 11, 14

Mean of Stock A:

$$\bar X=\frac{2+3+5+4+6}{5}=4$$

Mean of Stock B:

$$\bar Y=\frac{5+8+10+11+14}{5}=9.6$$

Using the sample covariance formula:

$$s_{XY}=\frac{\sum(X_i-\bar X)(Y_i-\bar Y)}{n-1}$$

A	B	$(A-4)$	$(B-9.6)$	Product
2	5	−2	−4.6	9.2
3	8	−1	−1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8
			Total	20

$$s_{XY}=\frac{20}{5-1}=\boxed{5}$$

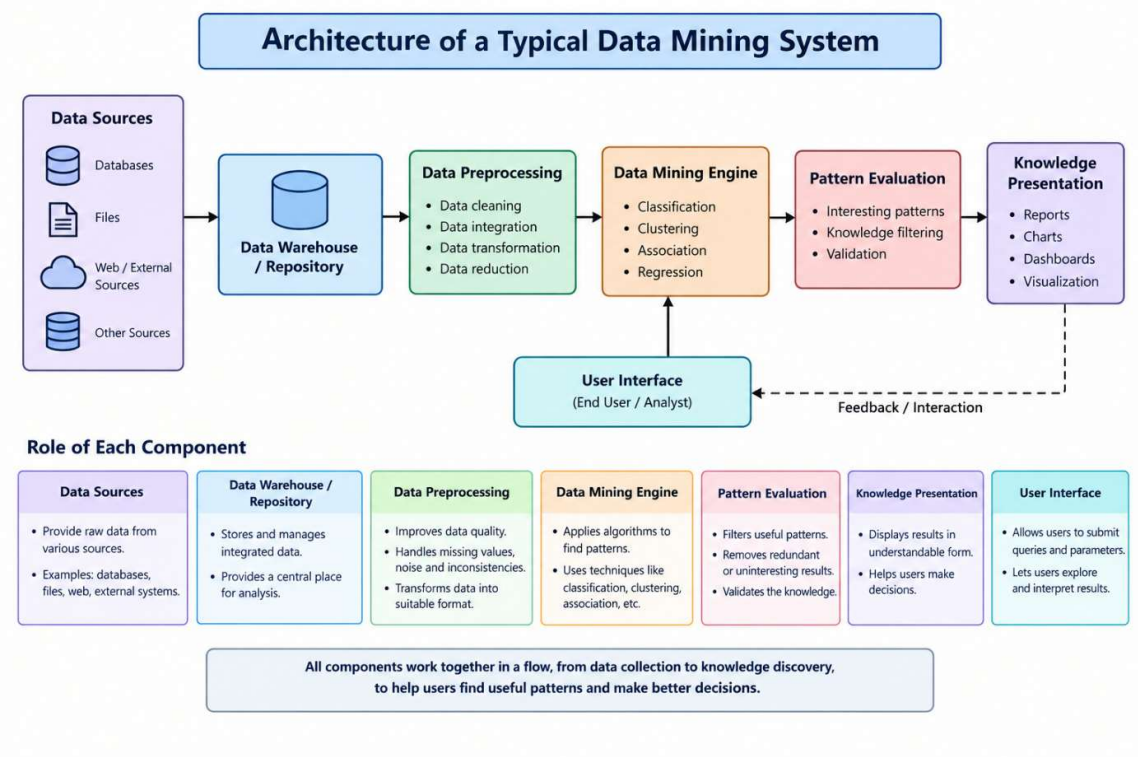
Interpretation: The covariance is positive, indicating that the two stocks tend to move in the same direction. Their prices are likely to rise or fall together rather than behave independently. However, positive covariance alone does not prove that they are always affected by the same industry trends.

Q6.)

Answer:

A data mining system discovers useful patterns and knowledge from large datasets using preprocessing techniques, mining algorithms, and pattern evaluation.

Architecture diagram:



Components and their roles:

1. Data Sources: Provide raw data from databases, files, and other systems.
2. Database/Data Warehouse Server: Retrieves and manages the required data.
3. Data Preprocessing: Cleans missing values, removes duplicates, and transforms data into a suitable format.
4. Data Mining Engine: Applies algorithms to identify patterns and relationships.
5. Pattern Evaluation: Identifies useful and meaningful patterns.
6. Knowledge Presentation: Displays results through reports, graphs, and visualizations.
7. User Interface: Allows users to interact with the system and interpret discovered knowledge.

How the components interact: Data is collected from sources, prepared by preprocessing, and analysed by the mining engine. Pattern evaluation filters useful

results, which are then presented to users. This process supports informed decisions and knowledge discovery.

Q7.)

Answer:

Given:

$$ X=\{4,5,6,7,9\},\quad Y=\{8,3,7,5,6\} $$

Step 1: Calculate the means

$$ \bar X=\frac{4+5+6+7+9}{5}=\frac{31}{5}=6.2 $$
$$ \bar Y=\frac{8+3+7+5+6}{5}=\frac{29}{5}=5.8 $$

Step 2: Calculate the deviations and products

X	Y	$X-6.2$	$Y-5.8$	Product
4	8	−2.2	2.2	−4.84
5	3	−1.2	−2.8	3.36
6	7	−0.2	1.2	−0.24
7	5	0.8	−0.8	−0.64
9	6	2.8	0.2	0.56
			Total	−1.80

Step 3: Calculate sample covariance

$$ s_{XY}=\frac{\sum(X_i-\bar X)(Y_i-\bar Y)}{n-1} $$
$$ s_{XY}=\frac{-1.80}{5-1} =\boxed{-0.45} $$

Result: The sample covariance is −0.45. The negative value indicates that the variables tend to move in opposite directions, although the covariance alone does not measure the strength of their relationship in a scale-independent way.

Q8.)

Answer:

Data preprocessing is the process of preparing raw data for analysis by improving its quality, consistency, and suitability for data mining.

Major steps in data preprocessing:

1. Data Cleaning: Handle missing values, duplicate records, incorrect values, and noise.
2. Data Integration: Combine data from multiple sources into a unified dataset.
3. Data Transformation: Normalize, standardize, or convert data into suitable formats.
4. Data Reduction: Reduce the number of attributes or records while preserving useful information.
5. Data Discretization: Convert continuous values into categories or intervals.

Systematic preprocessing pipeline:

Step	Technique	Example
1. Data inspection	Identify quality problems	Find missing ages and incorrect dates
2. Missing-value handling	Median or mean imputation	Replace a missing age with the median
3. Outlier treatment	Verify, cap, or remove invalid values	Investigate an age of 200
4. Format standardization	Apply consistent formats	Convert M, male, and MALE to Male
5. Data integration	Match records and remove duplicates	Combine customer records from two files

6. Transformation	Normalize numerical attributes	Scale spending values to 0–1
7. Data reduction	Select relevant features	Remove duplicate or irrelevant columns
8. Validation	Check the cleaned dataset	Verify missing values and data types

Justification of the sequence: First, the dataset is inspected to identify problems. Missing values and invalid observations are then handled, and formats are standardized. Data from multiple sources is integrated, followed by transformation and reduction. Finally, the prepared dataset is validated before mining.

This sequence reduces errors, prevents inconsistent values from affecting analysis, and improves the reliability of results.

Q9.)

Given ages:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Answer:

Equal-frequency binning, also called equi-depth binning, divides sorted data into bins containing approximately equal numbers of records. Local smoothing replaces individual values with representative values, such as bin means or medians.

Step 1: Divide the data into three bins

There are 27 age values.

\[\text{Records per bin}=\frac{27}{3}=9 \]

Step 2: Form the bins

Bin	Sorted ages	Bin mean
Bin 1	13, 15, 16, 16, 19, 20, 20, 21, 22	18.00
Bin 2	22, 25, 25, 25, 25, 30, 33, 33, 35	28.11
Bin 3	35, 35, 35, 36, 40, 45, 46, 52, 70	43.78

Step 3: Apply smoothing by bin means

Each age is replaced by the mean of its bin.

Bin	Original ages	Smoothed values
Bin 1	13, 15, 16, 16, 19, 20, 20, 21, 22	18.00 repeated 9 times
Bin 2	22, 25, 25, 25, 25, 30, 33, 33, 35	28.11 repeated 9 times
Bin 3	35, 35, 35, 36, 40, 45, 46, 52, 70	43.78 repeated 9 times

Result: The smoothed dataset consists of three representative values: 18.00, 28.11, and 43.78, each repeated nine times.

Conclusion: Equal-frequency binning reduces minor variations and simplifies age analysis. However, smoothing removes some individual-level detail, and the value 70 should be checked before treating it as noise because it may represent a valid patient age.

Q10.)

Answer:

Data discretization is the process of converting continuous numerical data into a limited number of intervals or categories. For example, ages can be grouped into Young, Middle-aged, and Senior categories.

Importance of data discretization:

1. Simplifies large numerical datasets.

2. Makes data easier to understand and interpret.
3. Reduces the effect of minor variations in some datasets.
4. Helps algorithms that work better with categorical attributes.
5. Supports reporting, classification, and decision-making.

Comparison of equal-width and equal-frequency binning

Feature	Equal-Width Binning	Equal-Frequency Binning
Meaning	Divides the data range into intervals of equal width	Divides sorted records into bins with approximately equal numbers of values
Basis	Minimum and maximum values	Number of records
Bin size	Equal numerical range	Approximately equal record count
Example	Ages 10–30, 30–50, 50–70	Three groups of nine patients from 27 ages
Advantage	Easy to calculate and interpret	Balances the number of records in each bin
Limitation	Some bins may contain very few records	Bin intervals can have very different widths

Example of equal-width binning:

Given ages 10, 20, 30, 40, 50, 60, 70, divide them into three equal-width bins:

- Bin 1: 10 to less than 30
- Bin 2: 30 to less than 50
- Bin 3: 50 to 70

Example of equal-frequency binning:

Using the same sorted ages, three approximately equal groups can be formed:

- Bin 1: 10, 20, 30
- Bin 2: 40, 50
- Bin 3: 60, 70

The groups contain 3, 2, and 2 records because seven records cannot be divided equally into three bins.

Impact on data analysis:

- Equal-width binning is useful when fixed numerical ranges have clear meaning, such as age groups or income ranges.
- Equal-frequency binning is useful when each group should contain a similar number of observations.
- Both methods simplify data, but the choice depends on the distribution and purpose of the analysis.
- Binning may hide important differences between individual values, so the original data should be preserved when needed.

Conclusion: Data discretization converts continuous values into meaningful groups. Equal-width binning maintains consistent interval sizes, whereas equal-frequency binning maintains approximately consistent record counts. Both techniques support data preprocessing and effective data mining.